Beyond Marginal Coverage: Evaluating Conformal Prediction Regions for Compositional (Sum-to-One) Outputs

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Abstract

Predictors whose output is a composition (a non-negative vector summing to one: softmax probabilities, topic or vote shares) are increasingly paired with conformal prediction regions in the simplex, but there is no shared recipe for judging whether such regions are *reliable*. We study this evaluation question with a controlled comparison of split conformal regions built from eight nonconformity scores (Euclidean, total variation, sup-norm, Hellinger, KL, Aitchison log-ratio, a Mahalanobis log-ratio score, a normalised Euclidean score), two Mondrian variants and cross-conformal calibration, on 36 synthetic settings (logistic-normal, Dirichlet and multinomial targets, $K \in \{3, 5, 10\}$, 200 replicates each) and on human label distributions for CIFAR-10 test images (CIFAR-10H). We find that (i) marginal coverage is uninformative: all scores are valid (mean coverage 0.895–0.911), and a single-run coverage check cannot resolve ± 0.02 unless the calibration set has about 2000 points, as the (known) Beta–Binomial law of one-run coverage, which we verify, predicts; (ii) region size depends strongly on the score's geometry and the noise family, with the Aitchison score giving regions 0.16–$0.87\times$ the Euclidean size for strictly positive logistic-normal targets but 1.8–$3.9\times$ for Dirichlet and multinomial targets with (near-)zeros; (iii) conditional coverage fails in score-specific ways: in closed form for $K{=}2$ the Euclidean score covers 0.83 to 0.997 of the targets at nominal 0.90, and on CIFAR-10H the Euclidean regions cover 0.99 and 0.72 of the lowest- and highest-entropy predictions, repaired by Mondrian calibration at $+23\%$ size; (iv) log-ratio scores depend on the zero-replacement constant ε: we prove that the threshold grows as $\sqrt{g/K}\,\log(1/\varepsilon)$ once about an α fraction of calibration compositions contain a zero, and confirm the slope to within 2%. We state the conditions under which these conclusions hold and where they should not be extrapolated.

1. Introduction

Many predictors output a composition $f(x) \in \Delta^{K-1} = \{p \in \mathbb{R}^K_{\geq 0} : \sum_k p_k = 1\}$: the softmax of a classifier, a topic mixture, a market share. Users need to know whether a stated uncertainty is trustworthy. Conformal prediction (Vovk et al., 2005; Lei et al., 2016; Angelopoulos & Bates, 2021) turns any score of disagreement $s(y, f(x))$ into a region $C(x) \subset \Delta^{K-1}$ with finite-sample marginal coverage under exchangeability, so the *validity* question has a formal answer, whatever the score. What remains open for compositional outputs is practical: which geometry to use for the score (Euclidean, Hellinger, KL, log-ratio (Aitchison, 1982)), how to handle exact zeros, which calibration scheme to use, and—the focus here—*how to evaluate* such regions in a way that is informative rather than circular.

We ask four questions, formulated before running the main experiments: **RQ1** What can empirical marginal coverage tell us, and how much calibration/test data does a coverage check need? **RQ2** How do region sizes of different geometries compare, and does the answer depend on the noise family and on how "size" is measured? **RQ3** Where does conditional coverage fail, which scores or calibrations repair it, and at what cost? **RQ4** How do log-ratio scores behave when targets contain exact zeros? Our a-priori expectations were: marginal coverage does not discriminate (H1); the best geometry follows the noise geometry (H2); the ranking of scores changes with the size measure (H3); log-ratio scores are fragile with zeros (H4). H1, H2, H4 are supported; H3 is only partly supported (Sec. 6.2).

Contributions. (1) An evaluation protocol for simplex-valued conformal regions combining the exact coverage law of a single run, group-conditional coverage gaps, size under two reference measures, and sensitivity to the zero-replacement constant. (2) Three propositions with derivations and numerical checks: the Beta–Binomial law of one-

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run coverage (a consequence of known results (Vovk, 2012), used here as an evaluation tool); a pivotality condition for conditional coverage with a closed form for $K{=}2$; and an exact asymptotic law for the log-ratio threshold under zero replacement. (3) A reproducible empirical study (36 synthetic cells $\times$ 200 replicates, plus CIFAR-10H) with uncertainty quantification. (4) Explicit conditions of validity and non-extrapolation (Sec. 7).

2. Related Work

Conformal prediction. Split/inductive conformal prediction (Papadopoulos et al., 2002; Vovk et al., 2005; Lei et al., 2016) gives marginal coverage for any score; its coverage given the calibration set is Beta distributed (Vovk, 2012), which we use as an evaluation tool. Jackknife+ and cross-conformal calibration (Barber et al., 2019; Vovk, 2015) reuse data at the price of a weaker $(1-2\alpha)$ guarantee; we test it at equal labelled budget. Exact conditional coverage is impossible in a distribution-free sense (Vovk, 2012; Barber et al., 2021); relaxations include Mondrian/class-conditional calibration (Vovk et al., 2005; Ding et al., 2023), multivalid methods (Jung et al., 2022) and conditional guarantees over function classes (Gibbs et al., 2023). Coverage under covariate shift (Tibshirani et al., 2019) is outside our scope.

Probability-vector outputs. For classifiers, conformal methods build label sets from softmax scores (Romano et al., 2020; Angelopoulos et al., 2020); Javanmardi et al. (2024) predict credal sets (sets of distributions), and Cresswell et al. (2024) study the downstream effect of prediction sets on decisions. Multi-target conformal regions in $\mathbb{R}^d$ use box, ellipsoid or copula-based scores (Messoudi et al., 2021). We instead study regions *in the simplex* for a compositional target, and compare geometries that respect (Aitchison) or ignore (Euclidean) the simplex constraint.

Compositional data analysis. Aitchison geometry, the centred and isometric log-ratio transforms and the logistic-normal family (Aitchison, 1982; Aitchison & Shen, 1980; Egozcue et al., 2003) are the standard tools; their known difficulty with exact zeros motivates our fourth question. We do not claim a new method: our contribution is the evaluation analysis, its theory, and the empirical map of when each choice works.

3. Setting, Methods and Evaluation Protocol

Setting. Data (X, Y) with $Y \in \Delta^{K-1}$; a predictor $f : \mathcal{X} \to \operatorname{int}\Delta^{K-1}$ trained on separate data; n calibration pairs exchangeable with the test pair. A score $s(y, p) \geq 0$ defines the region $C(x) = \{y \in \Delta^{K-1} : s(y, f(x)) \leq \hat{q}\}$ with $\hat{q}$ the $\lceil (n+1)(1-\alpha)\rceil$-th smallest calibration score (∞ if it exceeds n); $\alpha = 0.1$ throughout.

Scores (all computed with a closed-form membership test, so regions need not be parametrised): Euclidean $\ell_2 = \|y - p\|_2$; total variation $\frac{1}{2}\|y - p\|_1$; ℓ_∞; Hellinger $\|\sqrt{y} - \sqrt{p}\|_2$; $\mathrm{KL}(y\|p)$; Aitchison $\|\mathrm{clr}(y_\varepsilon) - \mathrm{clr}(p)\|_2$ with $y_\varepsilon = (y + \varepsilon\mathbf{1})/(1 + K\varepsilon)$ (additive zero replacement, $\varepsilon = 10^{-3}$ synthetic, $10^{-2} \approx$ half a vote for CIFAR-10H); a Mahalanobis score on $\mathrm{ilr}(y_\varepsilon) - \mathrm{ilr}(p)$ with a residual covariance estimated on an auxiliary fold; and $\ell_2/\hat{\sigma}(p)$ where $\hat{\sigma}$ is a linear function of the Gini impurity $1 - \|p\|^2$ fitted on the auxiliary fold. Validity never depends on the auxiliary fold.

Calibration schemes. (a) Split conformal. (b) Mondrian: separate $\hat{q}$ in three terciles of the entropy of $f(x)$ (cut points from the auxiliary fold). (c) Five-fold cross-conformal (Vovk, 2015) generalised to arbitrary scores: $y \in C(x)$ iff $\#\{i : s(y, f_{-k(i)}(x)) > R_i\} < (1-\alpha)(N+1)$, with R_i the out-of-fold scores.

Evaluation protocol. For each replicate we report: *marginal coverage* on a test set; *group-conditional coverage*, pooled over replicates, in entropy terciles of $f(x)$ and in argmax-class groups, summarised by the *gap* = worst group $|\text{coverage} - (1-\alpha)|$ (ent.-gap, arg-gap; their noise floor is about 0.003–0.008); and *size* = $\mathbb{E}_x\,\mu(C(x))$ for two reference measures μ: the empirical marginal law of Y (*pool size*, from an independent pool of 1000 targets) and the uniform (Dirichlet(1)) measure on the simplex (*uniform size*, 4000 draws, $K \leq 5$). Uncertainty: R independent replicates (new world, data, splits), mean $\pm$ 95% normal CI over replicates, paired differences within replicate with Holm correction over all 252 size comparisons; group gaps have bootstrap CIs over replicates.

4. Theory

Proofs are in Appendix A; all three statements are checked numerically in Sec. 6.

Proposition 4.1 (One-run coverage law)**.** *Let f be fixed and the scores $S_1, \ldots, S_{n+1}$ i.i.d. with continuous cdf F. With $k = \lceil (n+1)(1-\alpha)\rceil$, $l = n+1-k = \lfloor (n+1)\alpha \rfloor$, the coverage given the calibration set is $C_{\mathrm{cal}} = F(\hat{q}) \sim \mathrm{Beta}(k, l)$, with mean $k/(n+1) \geq 1-\alpha$ and variance $kl/((n+1)^2(n+2)) \approx \alpha(1-\alpha)/(n+2)$. The number of covered points among m test points is $\mathrm{BetaBin}(m; k, l)$, so the empirical coverage has variance $\approx \alpha(1-\alpha)\,(\frac{1}{n+2} + \frac{1}{m})$.*

Consequence. A one-run check "$|\widehat{\mathrm{cov}} - 0.9| \leq \delta$" of a valid method fails with a probability computable from this law, *whatever the score*. For $\delta = 0.02$, $n = 500$ and $m = 10^4$ the false-alarm probability is 14%; it falls to 0.6% only at $n = 2000$, and is 50% or more at $n = 100$ for every m (Fig. 1b).

Proposition 4.2 (Pivotal scores and conditional coverage)**.** *(a) If $S = s(Y, f(X))$ is independent of X, then for ev-*

ery calibration set and every x, $\mathbb{P}(Y \in C(X) \mid X = x, D_{\text{cal}}) = F(\hat{q})$: conditional coverage is constant in x. (b) For the logistic-normal model $Y = \text{softmax}(\eta(X) + \sigma\xi)$, $\xi \sim N(0, I_K)$, and $f = \text{softmax}(\eta)$, the Aitchison score ($\varepsilon = 0$) equals $\sigma\|\xi - \bar{\xi}\mathbf{1}\| \sim \sigma\chi_{K-1}$, hence is pivotal. (c) For $K = 2$, put $u = \text{logit}\, f_1$, $\tau = \sigma\sqrt{2}$, and let the ℓ_2 region be $|y_1 - f_1| \le r$. Then

$$c(f_1) = \Phi\Big(\tfrac{\text{logit}(f_1+r)-u}{\tau}\Big) - \Phi\Big(\tfrac{\text{logit}(f_1-r)-u}{\tau}\Big),$$

with $\text{logit}(t) = \pm\infty$ for $t \ge 1$, $t \le 0$. c is continuous with $c(f_1) \to 1$ as $f_1 \to 0$ or 1; since its average over $f_1(X)$ equals $1 - \alpha$ for the population threshold, $c < 1 - \alpha$ on a set of positive probability: no Euclidean region can be conditionally valid under this noise.

Proposition 4.3 (Zero replacement and the log-ratio threshold)**.** *Let $y \in \Delta^{K-1}$ have exactly z zero entries, $0 < z < K$, $f \in \text{int}\,\Delta^{K-1}$ fixed, $L = \log(1/\varepsilon)$. Then $d_A(y_\varepsilon, f) = L\sqrt{z(K-z)/K} + O(1)$ as $\varepsilon \to 0$, while $d_A(y_\varepsilon, f) = O(1)$ if $z = 0$. Hence for a fixed calibration set with $g_i = z_i(K - z_i)$,*

$$\lim_{\varepsilon\to 0} \hat{q}/\log(1/\varepsilon) = \sqrt{g_{(k)}/K},$$

where $g_{(k)}$ is the k-th smallest g_i. If $g_{(k)} > 0$ (equivalently, at least $l \approx \alpha(n+1)$ calibration compositions contain a zero), then $\hat{q} \to \infty$ linearly in $\log(1/\varepsilon)$ and the uniform size of $C(x)$ tends to 1; if $g_{(k)} = 0$, $\hat{q}$ stays bounded.

The practical reading: the Aitchison score is safe as ε varies only when fewer than about an α fraction of targets contain zeros; otherwise ε is a hidden tuning parameter of the *evaluation* (it controls region size) rather than a numerical detail.

5. Experimental Setup

Synthetic data. Features $x \sim N(0, I_5)$; $\eta(x) = W^\top x$ with $W_{jk} \sim N(0, 0.6^2)$ redrawn per replicate; $m(x) = \text{softmax}(\eta)$. Targets: *LN-homo* $Y = \text{softmax}(\eta + 0.5\,\xi)$; *LN-hetero* noise scale $0.35\,e^{0.5x_1}$; *Dirichlet* $Y \sim \text{Dir}(20\,m(x))$; *Multinomial* $Y = \text{Mult}(20, m(x))/20$ (exact zeros in 64% of the K=5 targets). $K \in \{3, 5, 10\}$, $n_{\text{cal}} \in \{50, 200, 1000\}$, 200 replicates per cell (7200 runs). The predictor is a multinomial-logit regression with soft labels (cross-entropy, ridge 10^{-3}) fit on 1000 points; 500 auxiliary, 1000 test and 1000 pool points are independent. Extra runs vary the predictor sample size ($n_{\text{train}} \in \{30, \ldots, 3000\}$), the zero constant ε, and the calibration scheme (equal labelled budget $N \in \{60, 150, 400\}$; 100 replicates).

Real data. CIFAR-10H (Peterson et al., 2019): for the 10000 CIFAR-10 (Krizhevsky, 2009) test images, the proportions of ≈50 human annotators choosing each class ($K = 10$; 44% unanimous; 99.97% of vectors contain at least one zero, 8.1 zero entries on average). The predictor is a publicly available ResNet-20 trained on the CIFAR-10 training set only (Chen, 2021) (92.1% test accuracy), used either as the raw softmax or after temperature scaling (Guo et al., 2017) fit on the auxiliary fold. Each replicate draws random disjoint auxiliary (1000), calibration ($n \in \{100, 500, 2000\}$) and test (2000) subsets of the 10000 images; 100 replicates. Intervals therefore reflect split-to-split variability within a fixed finite pool, not sampling of new images.

6. Results

6.1. Marginal coverage is valid and uninformative (RQ1)

Across all 36 cells, every non-Mondrian score has mean coverage between 0.895 and 0.911 (Table 3, App. B); the deviation from the exact expectation $k/(n+1)$ has $|z| \le 2.96$ for seven scores, and 3.41 for one of 288 score-cell pairs for Aitchison (Bonferroni threshold ≈ 3.8), so no evidence against validity. The Beta law of Prop. 4.1 matches simulation closely (Fig. 1a, Table 4): means within 1.5×10^{-3}, standard deviations within 1.3%, KS statistics ≤ 0.013 for 2×10^4 draws, for the ℓ_2, Aitchison and (multinomial) TV scores alike. Mondrian regions are valid but conservative in small groups (mean coverage 0.928, 0.907, 0.901 at $n_{\text{cal}} = 50, 200, 1000$, as the rounding in k predicts). *Coverage therefore cannot discriminate between scores, and a single coverage number is a noisy check*: the standard deviation of one-run coverage is 0.041 at $n = 50$ and 0.013 at $n = 500$ (Fig. 1b).

6.2. Region size depends on geometry, noise family and measure (RQ2)

Table 1 and Fig. 2 show that no score dominates. For strictly positive logistic-normal targets the log-ratio scores give the smallest regions, and the advantage grows with K: Aitchison regions have 0.79, 0.45, 0.16× the ℓ_2 pool size at $K = 3, 5, 10$ (LN-homo, $n_{\text{cal}} = 200$) and 0.87, 0.61, 0.35× for heteroscedastic noise. For Dirichlet and multinomial targets the picture reverses: Aitchison is 1.8–2.5× larger than ℓ_2 (Dirichlet) and 2.8–3.9× (multinomial), all Holm-significant. Hellinger is smaller than ℓ_2 in 27 cells (all LN, LN-hetero, Dirichlet cells) but larger in the 9 multinomial cells; KL is the best or tied-best non-log-ratio score in the Dirichlet and multinomial settings (KL vs. Hellinger: 24 of 36 cells smaller, 4 larger, 8 n.s.). The crude normalised score $\ell_2/\hat{\sigma}$ never helps: it is larger than ℓ_2 in 24 cells (+4–6%) and indistinguishable in 12. With a weaker or stronger predictor ($n_{\text{train}} \in [30, 3000]$, K=5) the winners are unchanged: Aitchison for both LN families, KL or Hellinger (near-tied) for Dirichlet and KL for multinomial targets

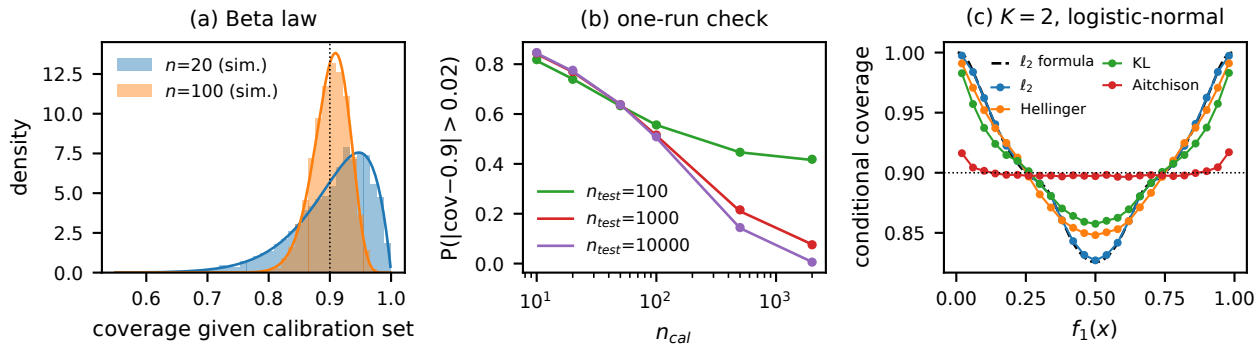


Figure 1. Evaluation theory vs. simulation. (a) Coverage given the calibration set (ℓ_2 score, logistic-normal, oracle f; 2×10^4 calibration draws per n) against the Beta law. (b) Probability that a valid method's one-run coverage lies more than 0.02 from 0.9; lines: Beta–Binomial law, dots: simulation. (c) K=2 conditional coverage vs. $f_1(x)$ ($\tau = 1$, $n_{\text{cal}} = 2000$, 40 calibrations); dashed: closed form of Prop. 4.2c.

Table 1. Synthetic results, $K = 5$, $n_{\text{cal}} = 200$, 200 replicates. *size*: mean pool size (fraction of reference targets inside the region; bold = smallest; 95% CI half-widths 0.004–0.013), *gap*: worst entropy-tercile coverage deviation $|\text{cov} - 0.9|$ (95% bootstrap half-width ≤ 0.006). Coverage of every row is within 0.898–0.908 (App. B). K=3, 10 and other n_{cal} in App. B.

	LN-homo		LN-hetero		Dirichlet		Multinom.	
Score	size	gap	size	gap	size	gap	size	gap
ℓ_2	0.149	0.026	0.125	0.020	0.108	0.037	0.114	0.042
TV	0.128	0.043	0.109	0.024	0.103	0.055	0.111	0.055
ℓ_∞	0.179	0.030	0.149	0.022	0.129	0.035	0.132	0.035
Hellinger	0.088	0.042	0.082	0.016	0.084	0.030	0.144	0.021
KL	0.088	0.041	0.080	0.016	**0.084**	0.024	**0.101**	0.028
Aitchison	**0.067**	0.003	**0.076**	0.030	0.272	0.086	0.443	0.018
Mahal.-ilr	0.067	0.004	0.077	0.031	0.280	0.090	0.452	0.016
$\ell_2/\hat{\sigma}$	0.160	0.040	0.133	0.034	0.115	0.024	0.120	0.021
ℓ_2-Mond.	0.158	0.008	0.136	0.007	0.116	0.008	0.121	0.006
Ait.-Mond.	0.071	0.007	0.081	0.010	0.241	0.007	0.451	0.013

(App. B).

Does the size measure matter? We expected the ranking to depend on the reference measure (H3). Over the 24 cells with $K \leq 5$, pool and uniform sizes give Kendall τ between methods with median 0.93 (min 0.64), so the overall ranking is largely stable; but the *best* method differs in 11/24 cells, always among near-ties (e.g. Aitchison vs. KL in LN-homo, K=3: pool sizes 0.79 and 0.82 of ℓ_2; Table 7, App.). Hence "which of two close scores is smaller" should not be reported under a single measure; the large effects above (factors 1.5–4) are measure-robust.

6.3. Conditional coverage fails in score-specific ways (RQ3)

All scores share marginal coverage ≈ 0.9 but differ in coverage across predicted-entropy terciles (Fig. 3, gap columns of Table 1). In the logistic-normal model the Aitchison score is near-pivotal (Prop. 4.2b is exact for the oracle centre $f = \text{softmax}(\eta)$; here f is the fitted regression): gap 0.003 (CI 0.002–0.007) versus 0.026 for ℓ_2 and 0.042 for Hellinger. This advantage is specific to the noise family: for Dirichlet targets the Aitchison gap is the *worst* (0.086 vs. 0.037), and with heteroscedastic noise it is 0.030. The K=2 closed form of Prop. 4.2c is confirmed (Fig. 1c): the Euclidean score covers 0.827 at $f_1 = 0.5$ and 0.997 near the boundary, the closed form deviating from simulation by at most 0.002, while Aitchison is flat (0.897–0.900 for $0.1 < f_1 < 0.9$; it rises to 0.917 at $f_1 = 0.02$ because the replacement $\varepsilon = 10^{-3}$ breaks pivotality). The worst coverage of the closed form is 0.80, 0.83, 0.88 for $\tau = 0.5, 1, 1.5$ (App.).

Mondrian calibration on entropy terciles removes the entropy gap for both scores (gap 0.006–0.013, about the rounding floor), but it is not free: for ℓ_2 it enlarges regions by 47%, 6%, 0.5% at $n_{\text{cal}} = 50, 200, 1000$ (averages over cells), and since the grouping is on the same variable on which we measure the gap this favours Mondrian; it does not repair other groupings: at $K = 10$ the argmax-class gap of ℓ_2 and ℓ_2-Mondrian are both 0.087 (LN-homo) whereas Aitchison reaches 0.014 in that setting.

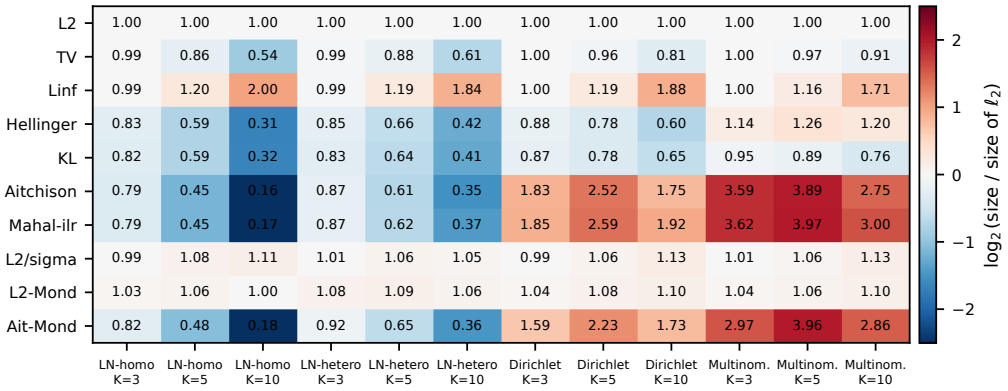


Figure 2. Pool size relative to ℓ_2 ($n_{\text{cal}} = 200$; entries are ratios; blue: smaller than ℓ_2, red: larger).

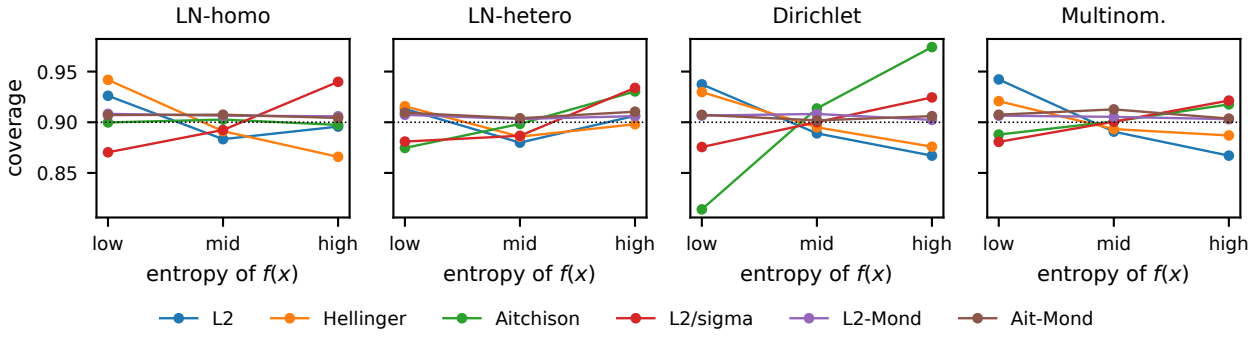


Figure 3. Conditional coverage by entropy tercile of $f(x)$ ($K = 5$, $n_{\text{cal}} = 200$, pooled over 200 replicates; dotted: 0.9). All curves have marginal coverage ≈ 0.9.

6.4. Zeros make the log-ratio score hostage to ε (RQ4)

The slope of $\hat{q}$ against $\log(1/\varepsilon)$ measured between $\varepsilon = 10^{-8}$ and 10^{-10} agrees with Prop. 4.3 to within 1.6% (range 0.05–1.6%) in all six (K, N) settings (mean over 200 replicates; e.g. $K = 5, N = 20$: 1.088 ± 0.004 vs. 1.090; $K = 10, N = 20$: 1.555 ± 0.001 vs. 1.580; Table 6, App.). Both regimes of the proposition occur: for K=3, N=100, only 8% of targets contain a zero ($< \alpha$) and 65% of replicates have $g_{(k)} = 0$ with a bounded threshold. In the size experiments the multinomial uniform size reaches 1.000 for $\varepsilon \leq 10^{-6}$ and Dirichlet 0.94 at 10^{-8}, while for logistic-normal targets (no zeros) the size is flat for $\varepsilon \leq 10^{-4}$. The size is non-monotone in ε (large ε distorts the data), and the best ε on our grid depends on the noise family (10^{-3} for LN, 10^{-2} for Dirichlet/multinomial). Even at its best grid value the Aitchison pool size is 0.138 (Dirichlet) and 0.244 (multinomial) vs. 0.084 (KL) and 0.101 (KL) (K=5, $n_{\text{cal}} = 200$); the ε choice is moreover made with knowledge of region size, i.e. optimistic.

6.5. Calibration scheme and predictor quality

At equal labelled budget, five-fold cross-conformal regions are smaller than split regions by 15%, 9%, 1% for $N = 60, 150, 400$ (mean over 4 noise families $\times$ ℓ_2/Hellinger/Aitchison; 12 cells each; Table 9, App.), with coverage 0.899–0.918 (the theoretical guarantee is $1 - 2\alpha$; we do not observe violations, but we test only $K = 5$ and three scores). The gain is concentrated at $N \lesssim 150$.

6.6. Real data: CIFAR-10H

Marginal coverage is 0.893–0.920 for all methods (Table 2), but conditional coverage is badly non-uniform: for ℓ_2 regions the coverage is 0.994, 0.979, 0.718 in the low, mid and high-entropy terciles of the raw softmax (ent.-gap 0.182, CI 0.175–0.187); Hellinger, TV and ℓ_∞ behave alike. Mondrian calibration brings the gap to 0.005 at a size cost of +23% (0.116 vs. 0.094); the argmax-class gap is not repaired (0.140). The log-ratio score is 5.2$\times$ larger than ℓ_2 for the raw softmax (0.489 vs. 0.094) and 1.2$\times$ after temperature scaling (0.113 vs. 0.094). With ε varied (Fig. 4c, Table 11), the size falls from 0.743 ($\varepsilon = 0.1$) to 0.126 (10^{-6}) then rises again (0.158 at 10^{-8}), as Prop. 4.3 pre-

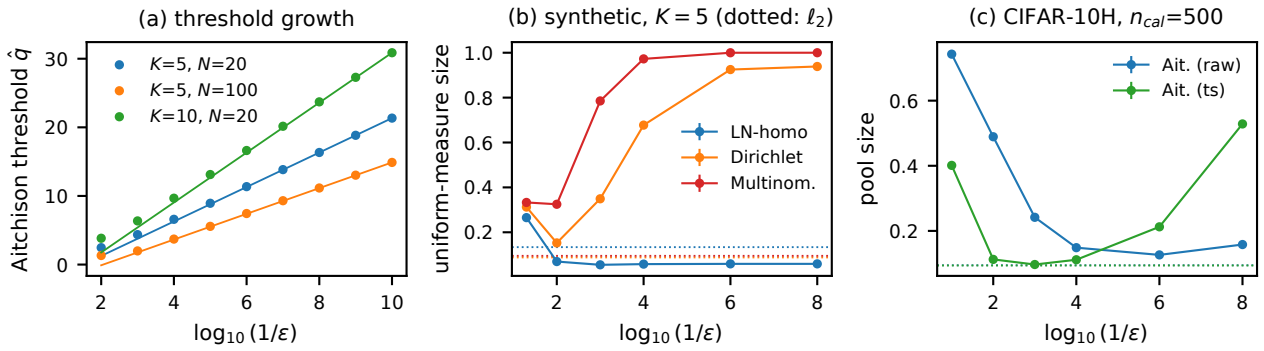


Figure 4. Zero-replacement sensitivity of the Aitchison score. (a) Mean threshold $\hat{q}$ vs. $\log_{10}(1/\varepsilon)$ for multinomial targets (200 replicates); lines: slope $\sqrt{g_{(k)}/K}$ of Prop. 4.3, anchored at the smallest ε. (b) Uniform-measure size, K=5, $n_{\text{cal}} = 200$ (dotted: ℓ_2). (c) CIFAR-10H pool size ($n_{\text{cal}} = 500$; dotted: ℓ_2).

Table 2. CIFAR-10H, $n_{\text{cal}} = 500$, 100 replicates. cov.: marginal coverage; size: pool size (bold: smallest); ent.-gap and arg-gap: worst entropy-tercile / argmax-class deviation from 0.9. 95% CI half-widths: coverage ≤ 0.003, size ≤ 0.006, ent.-gap and arg-gap ≤ 0.008.

	raw softmax				temperature-scaled			
Score	cov.	size	ent.-gap	arg-gap	cov.	size	ent.-gap	arg-gap
ℓ_2	0.897	0.094	0.182	0.092	0.899	0.094	0.182	0.074
TV	0.897	0.095	0.182	0.094	0.898	0.094	0.187	0.079
ℓ_∞	0.897	0.095	0.180	0.092	0.898	0.094	0.186	0.075
Hellinger	0.898	**0.094**	0.176	0.089	0.897	0.092	0.191	0.082
KL	0.898	0.095	0.108	0.134	0.898	**0.092**	0.160	0.078
Aitchison	0.903	0.489	0.176	0.092	0.902	0.113	0.085	0.092
Mahal.-ilr	0.904	0.484	0.177	0.036	0.901	0.106	0.079	0.053
$\ell_2/\hat{\sigma}$	0.899	0.134	0.092	0.126	0.900	0.131	0.019	0.152
ℓ_2-Mond.	0.902	0.116	0.005	0.140	0.901	0.105	0.005	0.136
Ait.-Mond.	0.906	0.295	0.009	0.083	0.903	0.109	0.005	0.082

dicts once ε is below the smallest predicted probabilities. For the raw softmax no ε reaches the Euclidean size (best 0.126 vs. 0.094); after temperature scaling the best value ($\varepsilon = 10^{-3}$) gives 0.097, comparable to ℓ_2, but the size rises to 0.213 and 0.529 at $\varepsilon = 10^{-6}, 10^{-8}$. The pool size saturates here (all Euclidean-type scores give 0.094–0.095): with 92% accuracy the best any region can do under this measure is roughly the share of one class (≈ 0.1), so size is uninformative among them, and conditional coverage is the discriminating metric. The normalised score $\ell_2/\hat{\sigma}$ halves the gap for the raw softmax (0.092) and nearly removes it after temperature scaling (0.019), at +43% size.

7. Discussion, Conditions and Limitations

What to report. Our results suggest the following minimum for compositional conformal regions: (1) marginal coverage with its Beta–Binomial reference band rather than a bare number, and $n_{\text{cal}} \gtrsim 2000$ if a ± 0.02 check is intended; (2) coverage in groups defined by the predictor (entropy, dominant component) with a stated noise floor; (3) size under at least two measures, with the observation that factor-level differences are robust but close pairs are not; (4) the zero fraction $\hat{\pi}_0$ of calibration targets, compared with α, and a sweep over ε for any log-ratio score.

Conditions of validity of the conclusions. The size ranking (Aitchison $\ll$ Hellinger/KL $<$ ℓ_2 for logistic-normal noise; reversed for Dirichlet/multinomial noise with small shape parameters/zeros) was observed for $x \in \mathbb{R}^5$ Gaussian features, a softmax-linear mean, one noise level per family, $K \leq 10$, $\alpha = 0.1$, a well-specified-ish predictor, and exchangeable data; Prop. 4.3 holds for additive zero replacement only. Prop. 4.2b needs the log-linear location structure and a predictor equal to the geometric centre. Prop. 4.1 needs exchangeability and no ties.

Where not to extrapolate. (i) Other noise levels, mean functions, $K \gg 10$, other α (the group-gap and size numbers are for $\alpha = 0.1$). (ii) Other zero treatments (multiplicative or model-based imputation) or other ε rules (e.g. ε tied to the number of counts). (iii) Covariate shift or temporal dependence: coverage guarantees and our diagnostics assume exchangeability. (iv) Real data beyond CIFAR-10H: this is one dataset, one predictor architecture, and replicates are resamples of a finite pool of 10000 images. (v) Class-conditional or label-set uses of softmax scores

(APS/RAPS-type sets), which we did not study. (vi) The conditional-coverage groups are defined from $f(x)$ only; coverage within groups defined by unobserved attributes can be different, and Mondrian on entropy does not address it.

Limitations. The normalised score uses a crude linear $\hat{\sigma}$; a better variance model may change its verdict. The Mahalanobis score uses a covariance that can be unstable at small auxiliary sizes (not varied). Cross-conformal was tested for three scores at $K{=}5$. The uniform measure is only evaluated for $K \leq 5$ (Monte Carlo in higher K is uninformative). The number of settings is large, but predictors, data-generating processes and sizes were chosen by us, and we did not tune any score except via the stated ε sweep. Hypothesis H3 (size ranking depends on measure) was only partially supported and we report it as such. An attempt to train the CIFAR predictor on CPU was abandoned because of compute limits; the pretrained model was used instead.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning, specifically the evaluation of uncertainty quantification. A potential negative effect is over-reliance on a region labelled "90%": our results show that this label refers to marginal coverage only and may hide large conditional undercoverage (e.g. 72% in the highest-entropy tercile on CIFAR-10H), which could matter if such regions are used in decisions about subgroups.

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A. Proofs

Proof of Proposition 4.1. Condition on f (trained on independent data). Since F is continuous and the data are i.i.d., $U_i = F(S_i)$ are i.i.d. $\mathrm{Unif}(0,1)$ (for merely exchangeable data only the marginal statement $\mathbb{E}C_{\mathrm{cal}} = k/(n+1)$ survives). The coverage given the calibration set is $\mathbb{P}(S_{n+1} \le \hat{q} \mid D_{\mathrm{cal}}) = F(\hat{q}) = U_{(k)}$, the k-th order statistic of n uniforms, which is $\mathrm{Beta}(k, n+1-k) = \mathrm{Beta}(k,l)$. Mean $k/(n+1)$ and variance $kl/((n+1)^2(n+2))$ are the Beta moments; $k = \lceil (n+1)(1-\alpha) \rceil = n+1-\lfloor (n+1)\alpha \rfloor$, so $l = \lfloor (n+1)\alpha \rfloor$ and $k/(n+1) \ge 1-\alpha$. Given D_{cal} the m test indicators are i.i.d. Bernoulli(C_{cal}), so their sum is $\mathrm{BetaBin}(m;k,l)$ and $\mathrm{Var}(\widehat{\mathrm{cov}}) = \mathrm{Var}(C_{\mathrm{cal}}) + \mathbb{E}[C_{\mathrm{cal}}(1-C_{\mathrm{cal}})]/m \approx \alpha(1-\alpha)(\frac{1}{n+2} + \frac{1}{m})$. □

Proof of Proposition 4.2. (a) If $S \perp X$ then $\mathbb{P}(S_{n+1} \le \hat{q} \mid X_{n+1} = x, D_{\mathrm{cal}}) = F(\hat{q})$ for every x, because $\hat{q}$ is a function of D_{cal} only. (b) With $f = \mathrm{softmax}(\eta)$, $\mathrm{clr}(f) = \eta - \bar{\eta}\mathbf{1}$ and $\mathrm{clr}(Y) = \eta - \bar{\eta}\mathbf{1} + \sigma(\xi - \bar{\xi}\mathbf{1})$, so $d_A(Y,f) = \sigma\|\xi - \bar{\xi}\mathbf{1}\|_2$, which is σ times the norm of the projection of a standard Gaussian onto a $(K-1)$-dimensional subspace, i.e. $\sigma\chi_{K-1}$ and independent of X. (c) For $K=2$, $Y_1 = \mathrm{expit}(\eta_1 - \eta_2 + \sigma(\xi_1 - \xi_2)) = \mathrm{expit}(u + \tau\zeta)$ with $\zeta \sim N(0,1)$, $\tau = \sigma\sqrt{2}$ and $u = \mathrm{logit}\, f_1$ for $f = \mathrm{softmax}(\eta)$. The ℓ_2 score is $\sqrt{2}|Y_1 - f_1|$, so $C = \{|y_1 - f_1| \le r\}$ with $r = \hat{q}/\sqrt{2}$, and, as expit is increasing, $c(f_1) = \mathbb{P}(\mathrm{logit}(f_1 - r) \le u + \tau\zeta \le \mathrm{logit}(f_1 + r))$, which is the displayed formula (with $\mathrm{logit}\, t = -\infty$ for $t \le 0$ and $+\infty$ for $t \ge 1$). Continuity is clear. As $f_1 \to 0$, $u \to -\infty$ and $\mathrm{logit}(f_1 + r) - u \to +\infty$ while the lower limit is $-\infty$ once $f_1 < r$, so $c \to 1$; symmetrically as $f_1 \to 1$. If $f_1(X)$ has support $(0,1)$ and $\mathbb{E}c(f_1(X)) = 1 - \alpha < 1$, then $c \ge 1-\alpha$ everywhere would force $c \equiv 1-\alpha$ almost surely and hence by continuity everywhere, contradicting $c \to 1$. Hence $c < 1-\alpha$ on a set of positive probability. □

Proof of Proposition 4.3. Let Z be the zero set of y, $|Z| = z$, and $L = \log(1/\varepsilon)$. For $i \in Z$, $\log y_{\varepsilon,i} = -L - \log(1 + K\varepsilon)$; for $j \notin Z$, $\log y_{\varepsilon,j} = \log(y_j + \varepsilon) - \log(1 + K\varepsilon) = O(1)$. The mean log is $-\frac{z}{K}L + O(1)$, so $\mathrm{clr}_i(y_\varepsilon) = -(1 - \frac{z}{K})L + O(1)$ for $i \in Z$ and $\mathrm{clr}_j(y_\varepsilon) = \frac{z}{K}L + O(1)$ for $j \notin Z$, while $\mathrm{clr}(f)$ is fixed. Therefore

$$d_A^2 = L^2\Big[z\big(1 - \tfrac{z}{K}\big)^2 + (K-z)\tfrac{z^2}{K^2}\Big] + O(L) = L^2\,\tfrac{z(K-z)}{K} + O(L),$$

and $d_A = L\sqrt{z(K-z)/K} + O(1)$ since $z(K-z) > 0$. If $z = 0$, $y_\varepsilon \to y$ and $d_A = O(1)$ (for $y \in \mathrm{int}\,\Delta^{K-1}$). Hence each calibration score satisfies $s_i/L \to \sqrt{g_i/K}$ and, order statistics being continuous, $\hat{q}/L = s_{(k)}/L \to \sqrt{g_{(k)}/K}$ (the order of the limits determines the k-th smallest). $g_{(k)} > 0$ iff at most $k-1$ calibration points have $g = 0$ iff at least $n - k + 1 = l$ points contain a zero. If $g_{(k)} > 0$, $\hat{q} \to \infty$, so for each fixed interior y, $d_A(y_\varepsilon, f) \le \hat{q}$ for ε small enough; by dominated convergence the uniform measure of $C(x)$ tends to that of the interior of the simplex, i.e. 1. If $g_{(k)} = 0$, $\hat{q}/L \to 0$ and in fact $\hat{q}$ stays bounded because scores of points with $z = 0$ are $O(1)$ and k-th smallest is attained among them. □

B. Additional Tables and Figures

Table 3. Validity of the eight non-Mondrian scores over all 36 synthetic cells (200 replicates each). z: deviation of mean coverage from the exact expectation $\lceil (n+1)(0.9) \rceil/(n+1)$ divided by its standard error over replicates.

Score	min cov.	max cov.	mean excess	max $\lvert z \rvert$	cells $\lvert z \rvert > 3$
ℓ_2	0.895	0.907	-0.0000	3.0	0/36
TV	0.896	0.907	+0.0001	2.5	0/36
ℓ_∞	0.896	0.907	+0.0000	2.5	0/36
Hellinger	0.898	0.909	+0.0008	2.6	0/36
KL	0.899	0.907	+0.0003	1.7	0/36
Aitchison	0.897	0.911	+0.0003	3.4	1/36
Mahal.-ilr	0.897	0.909	+0.0002	2.7	0/36
$\ell_2/\hat{\sigma}$	0.896	0.908	+0.0001	2.2	0/36

C. Reproducibility Details

Code, results (CSV) and a step-by-step README are provided with the submission (`code/`, `results/`, `README.md`). Software: Python 3.11, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6, Matplotlib 3.11.2, PyTorch 2.15.0 (CPU, inference only), PyArrow 26.0.1. Seeds: every synthetic run uses `np.random.SeedSequence([20261111, config_id,`

Table 4. Coverage given the calibration set vs. the Beta law of Prop. 4.1 (2×10^4 calibration sets per row; reference cdf from 4×10^6 draws; $\alpha = 0.1$, $K = 3$, oracle predictor).

DGP/score	n	mean (emp.)	mean (Beta)	sd (emp.)	sd (Beta)	5% q (emp.)	5% q (Beta)
LN / ℓ_2	10	0.9094	0.9091	0.0835	0.0830	0.740	0.741
LN / ℓ_2	20	0.9039	0.9048	0.0632	0.0626	0.783	0.784
LN / ℓ_2	50	0.9018	0.9020	0.0409	0.0412	0.827	0.826
LN / ℓ_2	100	0.9006	0.9010	0.0296	0.0296	0.848	0.848
LN / ℓ_2	500	0.9000	0.9002	0.0134	0.0134	0.877	0.877
LN / ℓ_2	2000	0.8999	0.9000	0.0067	0.0067	0.889	0.889
LN / Aitchison	10	0.9076	0.9091	0.0841	0.0830	0.738	0.741
LN / Aitchison	20	0.9046	0.9048	0.0625	0.0626	0.783	0.784
LN / Aitchison	50	0.9018	0.9020	0.0410	0.0412	0.827	0.826
LN / Aitchison	100	0.9011	0.9010	0.0295	0.0296	0.848	0.848
LN / Aitchison	500	0.9002	0.9002	0.0134	0.0134	0.877	0.877
LN / Aitchison	2000	0.9000	0.9000	0.0067	0.0067	0.889	0.889
Multinom. / TV	10	0.9091	0.9091	0.0823	0.0830	0.743	0.741
Multinom. / TV	20	0.9044	0.9048	0.0626	0.0626	0.784	0.784
Multinom. / TV	50	0.9023	0.9020	0.0410	0.0412	0.826	0.826
Multinom. / TV	100	0.9012	0.9010	0.0299	0.0296	0.848	0.848
Multinom. / TV	500	0.9004	0.9002	0.0134	0.0134	0.877	0.877
Multinom. / TV	2000	0.9002	0.9000	0.0068	0.0067	0.889	0.889

Table 5. Closed-form conditional coverage of the ℓ_2 region for $K = 2$ (Prop. 4.2c) with $u = \text{logit}\, f_1 \sim N(0, 1.5^2)$ and population threshold r solving marginal coverage 0.9.

τ	threshold r	min $c(f_1)$	f_1 at min	max $c(f_1)$
0.25	0.079	0.797	0.50	1.000
0.50	0.156	0.802	0.50	1.000
1.00	0.295	0.826	0.50	1.000
1.50	0.411	0.878	0.29	1.000
2.00	0.501	0.864	0.25	1.000

`replicate])` (CIFAR-10H: `[20261112, n_cal, replicate]`; cross-conformal: `[20261113, config_id, replicate]`; theory scripts: fixed seeds 1, 2, 3). Hardware: CPU only (shared 40-core machine under heavy load); wall-clock times observed with 30 worker processes: main grid 2030 s, predictor-quality grid 1390 s, ε grid 270 s, cross-conformal 430 s, CIFAR-10H 140–300 s, theory scripts about 10 min; figures and tables take under 5 minutes. No GPU was used.

Script to artefact map. `run_theory.py` → Fig. 1, Tables 4, 5, 6, Fig. 4a; `run_synth.py main` → Table 1, Figs. 2, 3, Tables 3, 7, 12, 13, 14; `run_synth.py predq` → Table 8; `run_synth.py eps` → Fig. 4b; `run_cv.py` → Table 9; `run_cifar10h.py` → Table 2, Figs. 4c, Tables 10, 11. Analysis: `analyze_synth.py`, `analyze_theory.py`, `analyze_aux.py`.

Data and licences. CIFAR-10 (Krizhevsky, 2009): no licence stated on the dataset page (citation requested); we used the `uoft-cs/cifar10` parquet mirror on Hugging Face (downloaded first because the official server was slow), then downloaded the official tarball (md5 `c58f30108f718f92721af3b95e74349a`, as listed on the task page) and verified that the mirror's test images and labels are identical, in the same order, and that its training images are identical as a set (`code/verify_cifar_mirror.py`); the test order also matches CIFAR-10H (argmax of human proportions equals the test label for 99.2% of images; a mismatched order would give about 10%). CIFAR-10H (Peterson et al., 2019): `cifar10h-probs.npy` from the authors' public repository; the repository states no licence in a LICENSE file (we checked); we use it for research only and do not redistribute it. The pretrained network is from Chen (2021) (`cifar10_resnet20`), loaded via `torch.hub`; it was trained on the CIFAR-10 training set only, so the 10000 test images are exchangeable and unseen.

Table 6. Slope of the Aitchison threshold $\hat{q}$ vs. $\log(1/\varepsilon)$ between $\varepsilon = 10^{-8}$ and 10^{-10} (multinomial targets, oracle $f = m(x)$, $n_{\mathrm{cal}} = 200$, 200 replicates) and the prediction $\sqrt{g_{(k)}/K}$ of Prop. 4.3; P(zero): fraction of targets with a zero.

K	N	P(zero)	slope (emp.)	slope (Prop. 3)	max |dev|
3	20	0.29	0.783 ± 0.022	0.784	5.6e-03
3	100	0.08	0.289 ± 0.054	0.290	1.5e-02
5	20	0.64	1.088 ± 0.004	1.090	5.2e-02
5	100	0.21	0.809 ± 0.041	0.814	1.9e-01
10	20	0.98	1.555 ± 0.001	1.580	5.4e-02
10	100	0.57	1.416 ± 0.012	1.420	1.7e-01

Table 7. Agreement of pool-size and uniform-size rankings over the ten methods ($K \leq 5$).

DGP	K	n_{cal}	Kendall τ	best (pool)	best (uniform)
LN-homo	3	50	0.86	Aitchison	KL
LN-homo	3	200	0.64	Aitchison	KL
LN-homo	3	1000	0.64	Aitchison	KL
LN-homo	5	50	1.00	Aitchison	Aitchison
LN-homo	5	200	1.00	Aitchison	Aitchison
LN-homo	5	1000	1.00	Aitchison	Aitchison
LN-hetero	3	50	0.95	KL	KL
LN-hetero	3	200	1.00	KL	KL
LN-hetero	3	1000	0.95	KL	KL
LN-hetero	5	50	0.87	Aitchison	KL
LN-hetero	5	200	0.73	Aitchison	KL
LN-hetero	5	1000	0.82	Ait.-Mond.	Ait.-Mond.
Dirichlet	3	50	1.00	KL	KL
Dirichlet	3	200	0.86	KL	KL
Dirichlet	3	1000	0.77	KL	KL
Dirichlet	5	50	0.96	Hellinger	KL
Dirichlet	5	200	0.96	KL	KL
Dirichlet	5	1000	1.00	KL	KL
Multinom.	3	50	0.91	KL	ℓ_2
Multinom.	3	200	0.82	KL	TV
Multinom.	3	1000	0.77	KL	ℓ_2
Multinom.	5	50	0.96	KL	KL
Multinom.	5	200	0.91	KL	TV
Multinom.	5	1000	0.96	KL	TV

Deviations and abandoned attempts. (1) A first attempt to train a ResNet-9-style network on CPU was abandoned (the shared machine was too loaded; no results from it are used). (2) The ICML 2026 template was used (the NeurIPS 2026 style file was not available online at the time of writing; ICML 2026 is the latest edition whose official template we could obtain). (3) The CIFAR-10H experiment uses 100 replicates instead of 200 for compute reasons. (4) Uniform-measure size is not computed for $K = 10$.

Literature access statement. Literature consisted of the starter list and a small number of well-known papers whose bibliographic data (title, authors, arXiv identifier, year) were verified through the arXiv API or Crossref by identifier or exact title. No search was made for, and we did not encounter, the specific published paper the task excludes; nothing was read from any page that cites or discusses it.

D. Checklist

- **Claims match results and scope:** yes; scope and non-extrapolation conditions are in Sec. 7.
- **Limitations discussed:** yes (Sec. 7); hypothesis H3 only partially supported and reported as such.
- **Theory: assumptions and proofs:** Props. 4.1–4.3, assumptions stated, proofs in App. A, each checked numerically.
- **Reproducibility:** code, seeds, commands, software versions, timings, script-to-table map (App. C, README).
- **Error bars / statistical significance:** 95% CIs over replicates, paired differences with Holm correction, bootstrap CIs

Table 8. Pool size vs. predictor training size ($K = 5$, $n_{\mathrm{cal}} = 200$, 200 replicates; bold: smallest among the six listed scores).

DGP	Score	n_{train}=30	n_{train}=100	n_{train}=300	n_{train}=3000
LN-homo	ℓ_2	0.200	0.166	0.159	0.152
	Hellinger	0.126	0.102	0.097	0.092
	KL	0.127	0.101	0.097	0.091
	Aitchison	**0.102**	**0.078**	**0.074**	**0.070**
	$\ell_2/\hat{\sigma}$	0.219	0.178	0.171	0.165
	ℓ_2-Mond.	0.209	0.175	0.171	0.162
LN-hetero	ℓ_2	0.172	0.136	0.132	0.128
	Hellinger	0.115	0.090	0.086	0.086
	KL	0.114	0.088	0.084	0.084
	Aitchison	**0.107**	**0.083**	**0.078**	**0.080**
	$\ell_2/\hat{\sigma}$	0.182	0.145	0.140	0.135
	ℓ_2-Mond.	0.181	0.149	0.147	0.142
Dirichlet	ℓ_2	0.153	0.117	0.110	0.107
	Hellinger	**0.114**	**0.090**	0.086	**0.083**
	KL	0.116	0.090	**0.085**	0.084
	Aitchison	0.284	0.271	0.268	0.268
	$\ell_2/\hat{\sigma}$	0.164	0.125	0.117	0.112
	ℓ_2-Mond.	0.162	0.126	0.119	0.114
Multinom.	ℓ_2	0.162	0.125	0.111	0.111
	Hellinger	0.166	0.150	0.138	0.139
	KL	**0.133**	**0.109**	**0.098**	**0.098**
	Aitchison	0.439	0.444	0.428	0.428
	$\ell_2/\hat{\sigma}$	0.173	0.133	0.117	0.116
	ℓ_2-Mond.	0.172	0.134	0.120	0.119

for group gaps.

- **Data licences / provenance:** documented above; no personal data beyond the public CIFAR-10H aggregate proportions.
- **Compute:** CPU only; no paid API.
- **Ethics / impact:** see Impact Statement.

Table 9. Cross-conformal (5-fold) vs. split conformal at equal labelled budget N ($K = 5$, 100 replicates; coverage of each; relative pool-size change of cross vs. split with 95% CI half-width in percentage points).

DGP	Score	N	cov. split	cov. cross	rel. size change
LN-homo	ℓ_2	60	0.907	0.909	-14% ± 4
		150	0.914	0.912	-9% ± 3
		400	0.899	0.904	-0% ± 2
	Hellinger	60	0.900	0.909	-15% ± 5
		150	0.913	0.911	-9% ± 3
		400	0.899	0.904	-1% ± 2
	Aitchison	60	0.907	0.910	-18% ± 5
		150	0.908	0.906	-8% ± 3
		400	0.900	0.903	-1% ± 2
LN-hetero	ℓ_2	60	0.909	0.912	-17% ± 7
		150	0.907	0.905	-13% ± 5
		400	0.901	0.903	-2% ± 3
	Hellinger	60	0.910	0.911	-20% ± 9
		150	0.910	0.905	-16% ± 5
		400	0.902	0.903	-1% ± 3
	Aitchison	60	0.906	0.908	-15% ± 8
		150	0.909	0.903	-16% ± 5
		400	0.904	0.904	-3% ± 4
Dirichlet	ℓ_2	60	0.901	0.914	-20% ± 7
		150	0.906	0.906	-9% ± 3
		400	0.903	0.904	-3% ± 2
	Hellinger	60	0.907	0.913	-18% ± 5
		150	0.909	0.906	-10% ± 3
		400	0.904	0.905	-3% ± 2
	Aitchison	60	0.898	0.904	-4% ± 4
		150	0.914	0.906	-4% ± 3
		400	0.903	0.902	-1% ± 1
Multinom.	ℓ_2	60	0.906	0.918	-20% ± 5
		150	0.911	0.909	-10% ± 3
		400	0.900	0.905	-1% ± 2
	Hellinger	60	0.906	0.907	-17% ± 7
		150	0.908	0.904	-6% ± 3
		400	0.898	0.903	+1% ± 2
	Aitchison	60	0.913	0.908	-4% ± 2
		150	0.905	0.899	-2% ± 1
		400	0.897	0.899	+0% ± 1

Table 10. CIFAR-10H pool size vs. calibration size.

	raw softmax			temperature-scaled		
Score	100	500	2000	100	500	2000
ℓ_2	0.097	0.094	0.095	0.096	0.094	0.094
TV	0.097	0.095	0.095	0.095	0.094	0.094
ℓ_∞	0.097	0.095	0.095	0.096	0.094	0.094
Hellinger	0.096	0.094	0.094	0.095	0.092	0.093
KL	0.097	0.095	0.095	0.094	0.092	0.092
Aitchison	0.482	0.489	0.485	0.114	0.113	0.112
Mahal.-ilr	0.471	0.484	0.475	0.107	0.106	0.105
$\ell_2/\hat{\sigma}$	0.141	0.134	0.139	0.141	0.131	0.129
ℓ_2-Mond.	0.145	0.116	0.113	0.121	0.105	0.104
Ait.-Mond.	0.356	0.295	0.281	0.118	0.109	0.108

Table 11. CIFAR-10H, raw softmax, $n_{\text{cal}} = 500$: Aitchison score for different ε (mean $\hat{q}$ and entropy-gap).

Score (ε)	cov.	size	$\hat{q}$	ent.-gap
Aitchison (10^{-8})	0.898	0.158	19.51	0.110
Aitchison (10^{-6})	0.898	0.126	14.26	0.086
Aitchison (10^{-4})	0.902	0.149	12.20	0.101
Aitchison (10^{-3})	0.902	0.242	13.03	0.160
Aitchison (10^{-2})	0.903	0.489	14.71	0.176
Aitchison (10^{-1})	0.903	0.743	16.61	0.179
ℓ_2	0.897	0.094	0.46	0.182
Hellinger	0.898	0.094	0.60	0.176

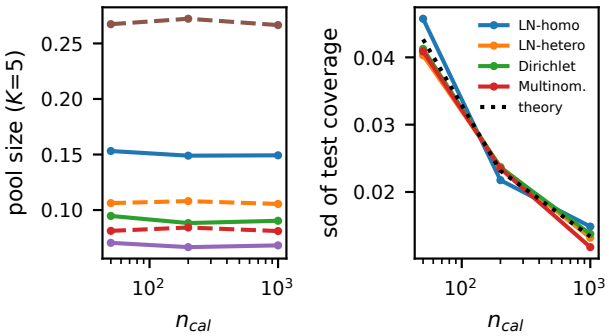


Figure 5. Left: pool size vs. n_{cal} ($K = 5$); right: standard deviation of test coverage across replicates for the ℓ_2 score ($n_{\text{test}} = 1000$) with the theoretical value $\sqrt{\alpha(1-\alpha)(1/(n+2)+1/m)}$.

Table 12. Paired relative size difference (first minus second, relative to second; $n_{\text{cal}} = 200$); *: Holm-adjusted $p < 0.05$ over 252 comparisons.

Comparison	K	LN-homo	LN-hetero	Dirichlet	Multinom.
Aitchison vs ℓ_2	3	-21%*	-13%*	+83%*	+259%*
	5	-55%*	-39%*	+152%*	+289%*
	10	-84%*	-65%*	+75%*	+175%*
Hellinger vs ℓ_2	3	-17%*	-15%*	-12%*	+14%*
	5	-41%*	-34%*	-22%*	+26%*
	10	-69%*	-58%*	-40%*	+20%*
Aitchison vs Hellinger	3	-5%*	+2%	+107%*	+213%*
	5	-25%*	-8%*	+223%*	+209%*
	10	-48%*	-17%*	+191%*	+129%*
KL vs Hellinger	3	-1%*	-2%*	-2%*	-17%*
	5	-0%	-3%*	-0%	-30%*
	10	+1%	-3%*	+8%*	-37%*
$\ell_2/\hat{\sigma}$ vs ℓ_2	3	-1%	+1%	-1%	+1%
	5	+8%*	+6%*	+6%*	+6%*
	10	+11%*	+5%*	+13%*	+13%*
ℓ_2-Mond. vs ℓ_2	3	+3%*	+8%*	+4%*	+4%*
	5	+6%*	+9%*	+8%*	+6%*
	10	+0%	+6%*	+10%*	+10%*
Ait.-Mond. vs Aitchison	3	+4%*	+6%*	-13%*	-17%*
	5	+6%*	+7%*	-12%*	+2%*
	10	+8%*	+5%	-1%	+4%*

Table 13. Pool size for the full synthetic grid (bold: smallest).

DGP	K	n_{cal}	ℓ_2	TV	ℓ_∞	Hellinger	KL	Aitchison	Mahal.-ilr	$\ell_2/\hat{\sigma}$	ℓ_2-Mond.	Ait.-Mond.
LN-homo	3	50	0.269	0.268	0.268	0.225	0.222	**0.215**	0.215	0.269	0.320	0.260
LN-homo	3	200	0.255	0.253	0.253	0.211	0.208	**0.202**	0.202	0.253	0.264	0.210
LN-homo	3	1000	0.255	0.252	0.252	0.211	0.209	**0.203**	0.203	0.253	0.255	0.205
LN-homo	5	50	0.153	0.133	0.185	0.095	0.094	**0.070**	0.071	0.165	0.215	0.105
LN-homo	5	200	0.149	0.128	0.179	0.088	0.088	**0.067**	0.067	0.160	0.158	0.071
LN-homo	5	1000	0.149	0.129	0.180	0.090	0.090	**0.068**	0.069	0.161	0.152	0.069
LN-homo	10	50	0.095	0.051	0.184	0.030	0.031	**0.016**	0.017	0.106	0.149	0.038
LN-homo	10	200	0.087	0.047	0.175	0.027	0.028	**0.014**	0.015	0.097	0.087	0.015
LN-homo	10	1000	0.083	0.046	0.168	0.026	0.027	**0.013**	0.014	0.093	0.079	0.014
LN-hetero	3	50	0.210	0.208	0.208	0.181	**0.179**	0.189	0.189	0.213	0.295	0.270
LN-hetero	3	200	0.212	0.209	0.209	0.180	**0.176**	0.184	0.185	0.214	0.228	0.195
LN-hetero	3	1000	0.209	0.206	0.206	0.177	**0.174**	0.180	0.180	0.209	0.212	0.179
LN-hetero	5	50	0.148	0.128	0.176	0.100	0.097	**0.093**	0.093	0.159	0.250	0.169
LN-hetero	5	200	0.125	0.109	0.149	0.082	0.080	**0.076**	0.077	0.133	0.136	0.081
LN-hetero	5	1000	0.124	0.109	0.148	0.082	0.080	0.076	0.076	0.131	0.126	**0.073**
LN-hetero	10	50	0.088	0.053	0.158	0.037	0.036	**0.029**	0.031	0.094	0.170	0.067
LN-hetero	10	200	0.076	0.046	0.140	0.032	0.031	**0.027**	0.028	0.080	0.081	0.028
LN-hetero	10	1000	0.072	0.043	0.134	0.029	0.028	0.024	0.026	0.076	0.068	**0.022**
Dirichlet	3	50	0.178	0.177	0.177	0.162	**0.157**	0.350	0.355	0.180	0.220	0.360
Dirichlet	3	200	0.175	0.175	0.175	0.155	**0.152**	0.321	0.323	0.174	0.181	0.278
Dirichlet	3	1000	0.172	0.172	0.172	0.154	**0.151**	0.325	0.328	0.172	0.173	0.271
Dirichlet	5	50	0.106	0.101	0.129	**0.081**	0.082	0.267	0.277	0.114	0.143	0.285
Dirichlet	5	200	0.108	0.103	0.129	0.084	**0.084**	0.272	0.280	0.115	0.116	0.241
Dirichlet	5	1000	0.105	0.101	0.126	0.081	**0.081**	0.267	0.276	0.111	0.108	0.226
Dirichlet	10	50	0.078	0.064	0.144	**0.046**	0.050	0.130	0.142	0.087	0.121	0.162
Dirichlet	10	200	0.076	0.062	0.144	**0.046**	0.050	0.134	0.146	0.086	0.084	0.132
Dirichlet	10	1000	0.072	0.059	0.136	**0.044**	0.047	0.130	0.141	0.082	0.075	0.123
Multinom.	3	50	0.183	0.184	0.184	0.210	**0.173**	0.620	0.625	0.186	0.235	0.581
Multinom.	3	200	0.176	0.176	0.176	0.202	**0.167**	0.632	0.637	0.178	0.183	0.523
Multinom.	3	1000	0.177	0.177	0.177	0.201	**0.168**	0.631	0.636	0.178	0.178	0.495
Multinom.	5	50	0.115	0.112	0.134	0.141	**0.099**	0.426	0.435	0.122	0.159	0.468
Multinom.	5	200	0.114	0.111	0.132	0.144	**0.101**	0.443	0.452	0.120	0.121	0.451
Multinom.	5	1000	0.108	0.106	0.125	0.136	**0.096**	0.427	0.437	0.114	0.111	0.429
Multinom.	10	50	0.079	0.072	0.136	0.097	**0.061**	0.221	0.240	0.088	0.130	0.281
Multinom.	10	200	0.075	0.068	0.129	0.091	**0.057**	0.207	0.226	0.085	0.083	0.215
Multinom.	10	1000	0.074	0.067	0.128	0.089	**0.056**	0.206	0.225	0.083	0.076	0.208

Table 14. Worst entropy-tercile coverage gap for the full synthetic grid.

DGP	K	n_{cal}	ℓ_2	TV	ℓ_∞	Hellinger	KL	Aitchison	Mahal.-ilr	$\ell_2/\hat{\sigma}$	ℓ_2-Mond.	Ait.-Mond.
LN-homo	3	50	0.059	0.064	0.064	0.057	0.054	0.009	0.008	0.023	0.026	0.031
LN-homo	3	200	0.058	0.063	0.063	0.058	0.055	0.003	0.003	0.022	0.008	0.009
LN-homo	3	1000	0.058	0.063	0.063	0.055	0.054	0.001	0.000	0.021	0.003	0.004
LN-homo	5	50	0.026	0.042	0.030	0.044	0.043	0.006	0.007	0.032	0.028	0.038
LN-homo	5	200	0.026	0.043	0.030	0.042	0.041	0.003	0.004	0.040	0.008	0.007
LN-homo	5	1000	0.025	0.042	0.029	0.042	0.040	0.003	0.003	0.039	0.003	0.002
LN-homo	10	50	0.034	0.020	0.031	0.028	0.028	0.013	0.016	0.051	0.028	0.029
LN-homo	10	200	0.036	0.019	0.034	0.025	0.025	0.020	0.018	0.054	0.010	0.010
LN-homo	10	1000	0.036	0.019	0.033	0.025	0.026	0.020	0.016	0.053	0.003	0.003
LN-hetero	3	50	0.042	0.045	0.045	0.030	0.030	0.023	0.024	0.018	0.028	0.028
LN-hetero	3	200	0.045	0.048	0.048	0.033	0.034	0.019	0.019	0.017	0.010	0.009
LN-hetero	3	1000	0.046	0.049	0.049	0.034	0.035	0.017	0.017	0.015	0.004	0.003
LN-hetero	5	50	0.017	0.024	0.016	0.017	0.017	0.036	0.035	0.040	0.036	0.038
LN-hetero	5	200	0.020	0.024	0.022	0.016	0.016	0.030	0.031	0.034	0.007	0.010
LN-hetero	5	1000	0.020	0.021	0.021	0.014	0.014	0.034	0.033	0.035	0.002	0.002
LN-hetero	10	50	0.045	0.027	0.045	0.028	0.027	0.053	0.050	0.053	0.037	0.029
LN-hetero	10	200	0.044	0.022	0.045	0.024	0.024	0.051	0.048	0.050	0.010	0.008
LN-hetero	10	1000	0.045	0.022	0.045	0.023	0.022	0.054	0.052	0.052	0.002	0.002
Dirichlet	3	50	0.055	0.058	0.058	0.027	0.025	0.104	0.102	0.017	0.033	0.033
Dirichlet	3	200	0.058	0.062	0.062	0.027	0.026	0.119	0.118	0.020	0.011	0.012
Dirichlet	3	1000	0.056	0.059	0.059	0.022	0.023	0.119	0.118	0.016	0.002	0.002
Dirichlet	5	50	0.039	0.055	0.037	0.029	0.024	0.086	0.087	0.025	0.024	0.030
Dirichlet	5	200	0.037	0.055	0.035	0.030	0.024	0.086	0.090	0.024	0.008	0.007
Dirichlet	5	1000	0.041	0.056	0.039	0.029	0.023	0.092	0.094	0.024	0.003	0.003
Dirichlet	10	50	0.024	0.063	0.013	0.046	0.034	0.071	0.069	0.031	0.033	0.033
Dirichlet	10	200	0.023	0.063	0.014	0.044	0.031	0.066	0.064	0.033	0.010	0.009
Dirichlet	10	1000	0.021	0.065	0.012	0.043	0.031	0.066	0.065	0.035	0.002	0.002
Multinom.	3	50	0.064	0.066	0.066	0.027	0.020	0.067	0.067	0.022	0.035	0.032
Multinom.	3	200	0.060	0.063	0.063	0.023	0.014	0.067	0.067	0.015	0.007	0.010
Multinom.	3	1000	0.062	0.064	0.064	0.023	0.017	0.066	0.066	0.016	0.004	0.004
Multinom.	5	50	0.049	0.061	0.043	0.025	0.033	0.017	0.017	0.030	0.036	0.029
Multinom.	5	200	0.042	0.055	0.035	0.021	0.028	0.018	0.016	0.021	0.006	0.013
Multinom.	5	1000	0.045	0.058	0.037	0.024	0.031	0.015	0.013	0.023	0.002	0.002
Multinom.	10	50	0.024	0.065	0.011	0.062	0.050	0.037	0.034	0.034	0.031	0.035
Multinom.	10	200	0.023	0.063	0.011	0.061	0.051	0.033	0.031	0.033	0.009	0.007
Multinom.	10	1000	0.022	0.063	0.011	0.060	0.049	0.035	0.031	0.037	0.002	0.003